

Point-wise

Reduction/Norm.

MatMul/Attention

Stencil/Conv.

Data Layout

B200

GH200

Triton

cuTile

TileLang

Triton

cuTile

TileLang

dequantize_rowwise

dropout

fused_activation

leaky_relu

mul2

quantize_global

relu

rope

sigmoid

swiglu

vector_add

weight_dequant

argmax

batch_normalization

cross_entropy

histogramming

kl_divergence

l2_norm

layernorm

mean_reduction

moe_topk_gating

rmsnorm

softmax

batched_matmul

block_sparse_attention

flash_attention

flash_decode

linear_self_attention

matmul_fp32_fp16_fp8

matmul_int8

streamk_matmul

1d_conv

2d_conv

2d_max_pooling

3d_conv

gaussian_blur

jacobi_stencil_2d

bitonic_sort

destindex

interleave

matrix_copy

matrix_transpose

radix_sort

reverse_array

top_k_selection

1.00	1.15	1.03
1.01	1.01	1.00
1.00	1.01	1.01
1.00	1.00	1.01
1.03	1.08	1.00
1.00	1.01	1.04
1.03	1.09	1.00
1.01	1.02	1.00
1.00	1.31	1.02
1.00	1.23	1.07
1.02	1.03	1.00
1.09	1.89	1.00
1.19	1.15	1.00
1.14	1.27	1.00
1.00	1.41	1.06
3.98	6.46	1.00
1.11	1.07	1.00
1.06	1.21	1.00
1.16	1.40	1.00
1.01	1.15	1.00
1.00	1.96	1.03
1.11	1.19	1.00
1.00	1.52	1.12
1.00	1.13	1.32
1.00	2.69	2.72
1.06	1.00	1.78
1.01	3.40	1.00
1.00	2.02	3.28
1.27	1.00	2.11
1.05	1.07	1.00
1.00	2.05	1.56
1.00	1.76	5.55
1.00	1.56	3.87
1.27	1.66	1.00
1.00	2.19	4.11
1.00	1.21	1.38
1.19	1.03	1.00
1.00	1.00	1.00
1.00	2.10	1.07
1.00	1.02	1.04
1.01	1.00	1.01
1.00	1.00	1.24
1.00	1.87	1.78
1.00	1.01	1.17
1.00	1.13	1.06

1.00	1.13	1.03
1.01	1.03	1.00
1.00	1.01	1.01
1.00	1.02	1.00
1.00	1.04	1.00
1.00	1.01	1.01
1.01	1.03	1.00
1.01	1.02	1.00
1.00	1.09	1.01
1.00	1.12	1.01
1.00	1.02	1.00
1.01	1.22	1.00
1.05	1.09	1.00
1.00	1.14	1.03
1.06	1.35	1.00
4.29	7.08	1.00
1.00	1.01	1.00
1.08	1.06	1.00
1.09	1.14	1.00
1.00	1.01	1.00
1.04	2.02	1.00
1.10	1.10	1.00
1.00	1.25	1.04
1.00	1.15	1.17
1.00	2.17	1.72
1.16	1.00	1.58
1.02	3.13	1.00
1.00	1.23	2.87
1.00	1.61	2.39
1.00	1.98	1.97
1.00	1.43	1.16
1.00	1.24	5.65
1.00	1.48	4.71
1.00	1.05	1.13
1.00	2.08	4.40
1.07	1.33	1.00
1.12	1.02	1.00
1.02	1.09	1.00
1.00	1.64	1.07
1.00	1.02	1.05
1.00	1.03	1.00
1.00	1.01	1.13
1.00	1.85	1.76
1.00	1.05	1.13
1.31	1.00	1.26

≥8×

4×

2×

1×

fastest

≤ 1.05×

slowdown vs. fastest DSL
on the same device